Long-term Unemployment Dynamics and Unemployment Insurance Extensions

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2 August 2018

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Introduction

- To what extent do unemployment insurance (UI) extensions explain
  1. The incidence of long-term unemployment (LTU)?
  2. The unemployment duration structure?
- Emphasis on the Great Recession in the US
  - With general applications to cyclical fluctuations
- Using equilibrium search and matching model with heterogeneous workers

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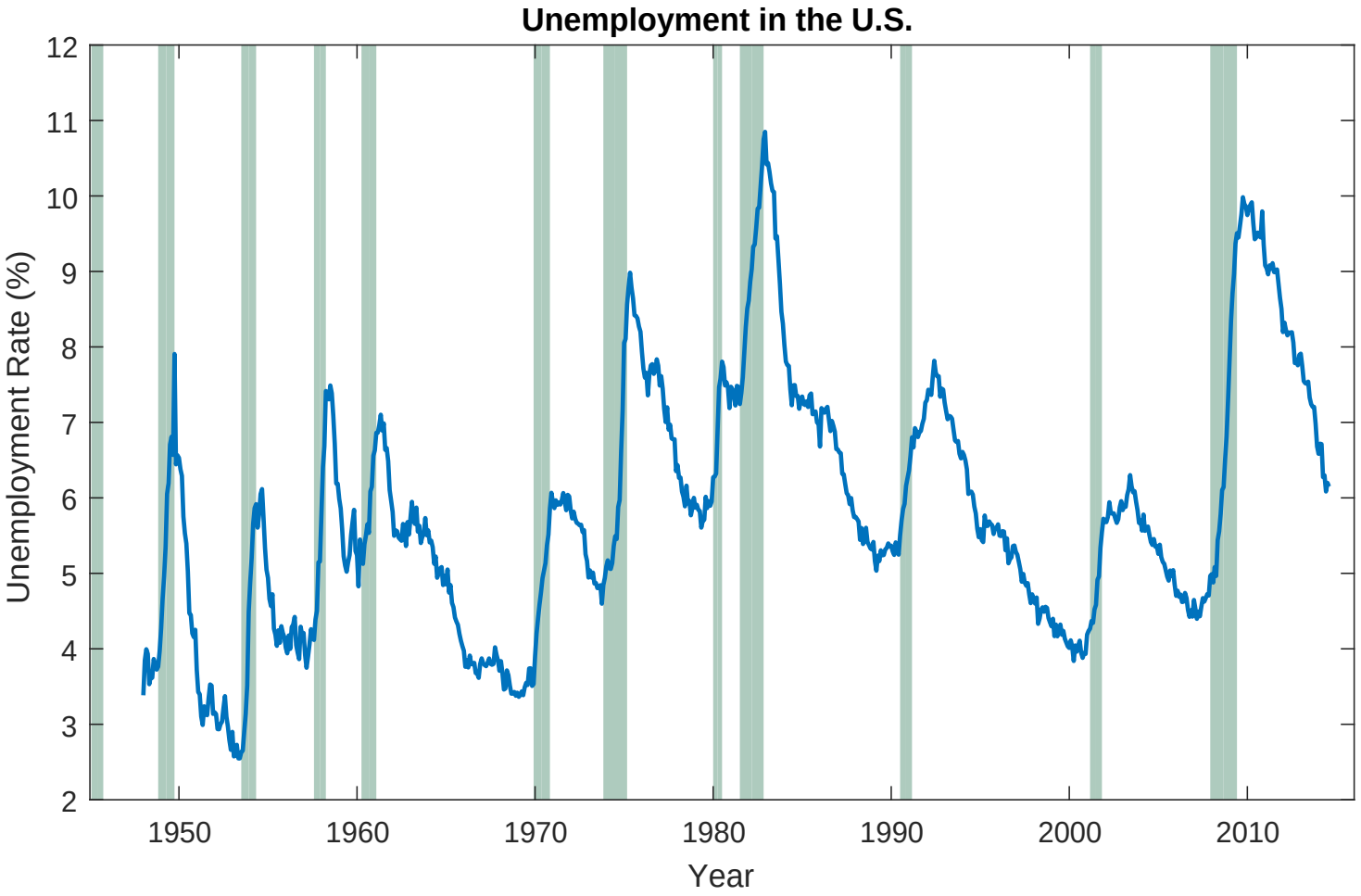
  - With general applications to cyclical fluctuations

- Using equilibrium search and matching model with heterogeneous workers

- It is well known that the long-term unemployed face worse labour market outcomes than the short-term unemployed
  - Understanding the sources of LTU is important especially for policy

Unemployment Rate in the US

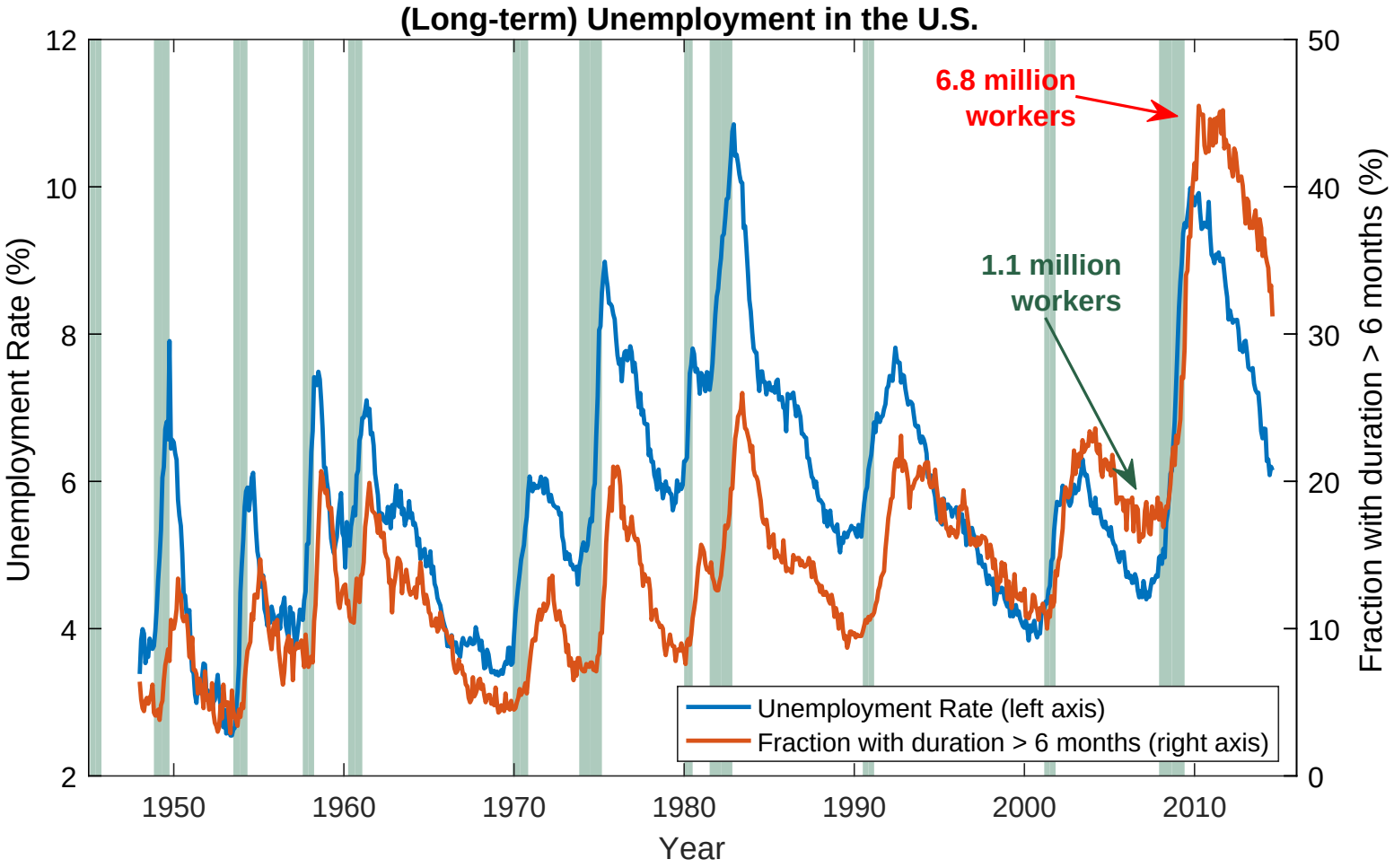
- Cyclical swings in unemployment, but the Great Recession is not the most severe



Source: CPS / Shaded areas denote recessions

Long-term Unemployment in the US

- Large unprecedented rise in the share of long-term unemployment in the Great Recession



Source: CPS / Shaded areas denote recessions ▸ LTU Rate

Introduction (cont.)

- I focus on **the role of UI extensions** on the dynamics of long-term unemployment and unemployment durations

- By making unemployment less unpleasant

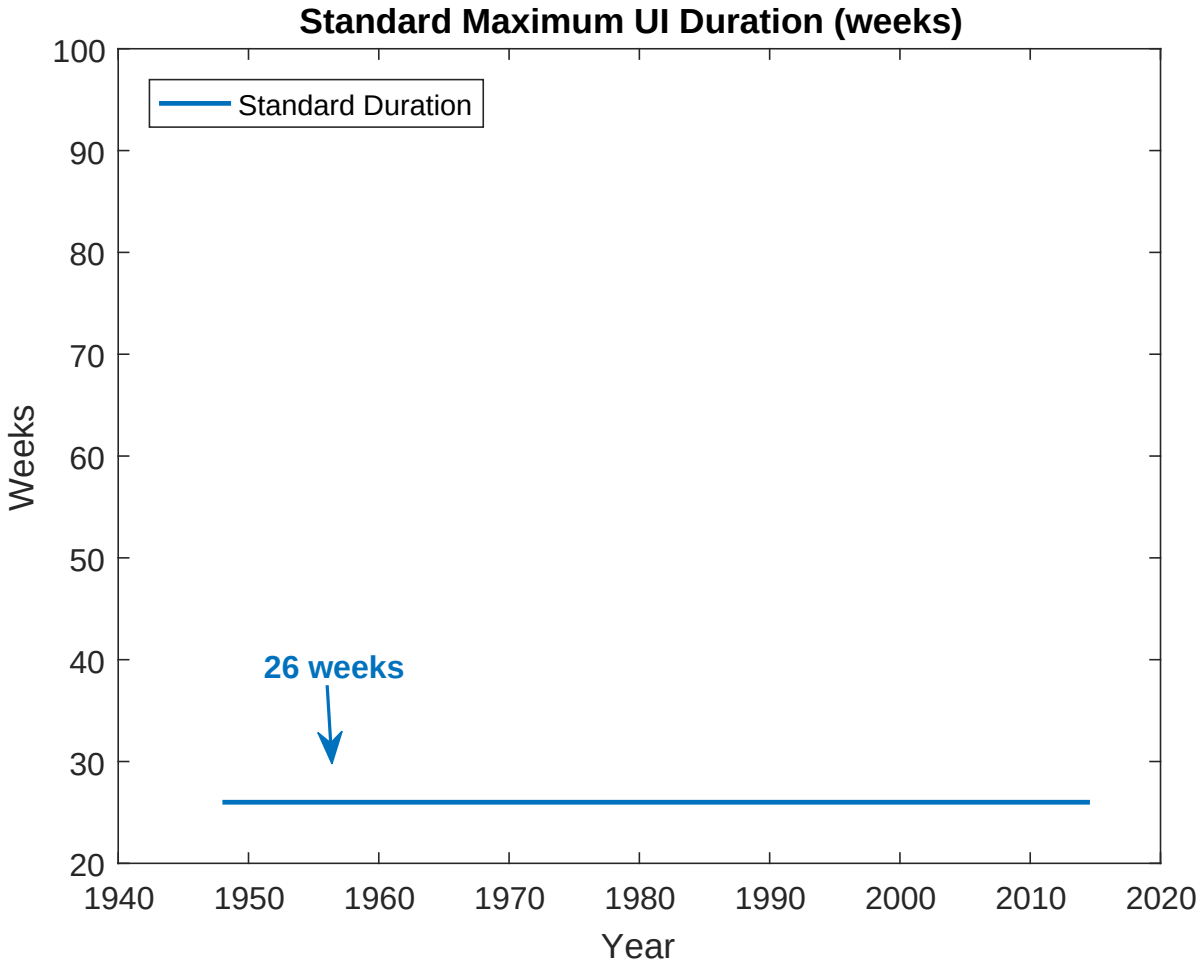
  1. **Reduce outflow** out of unemployment (micro)
     - Lower incentive for job search
     - More selective with job offers

  2. **Increase inflow** into unemployment (congestion, composition - macro)
     - Induce marginal workers to return to unemployment

- Huge and unprecedented rise in the maximum UI duration during the Great Recession

UI Extensions in the US

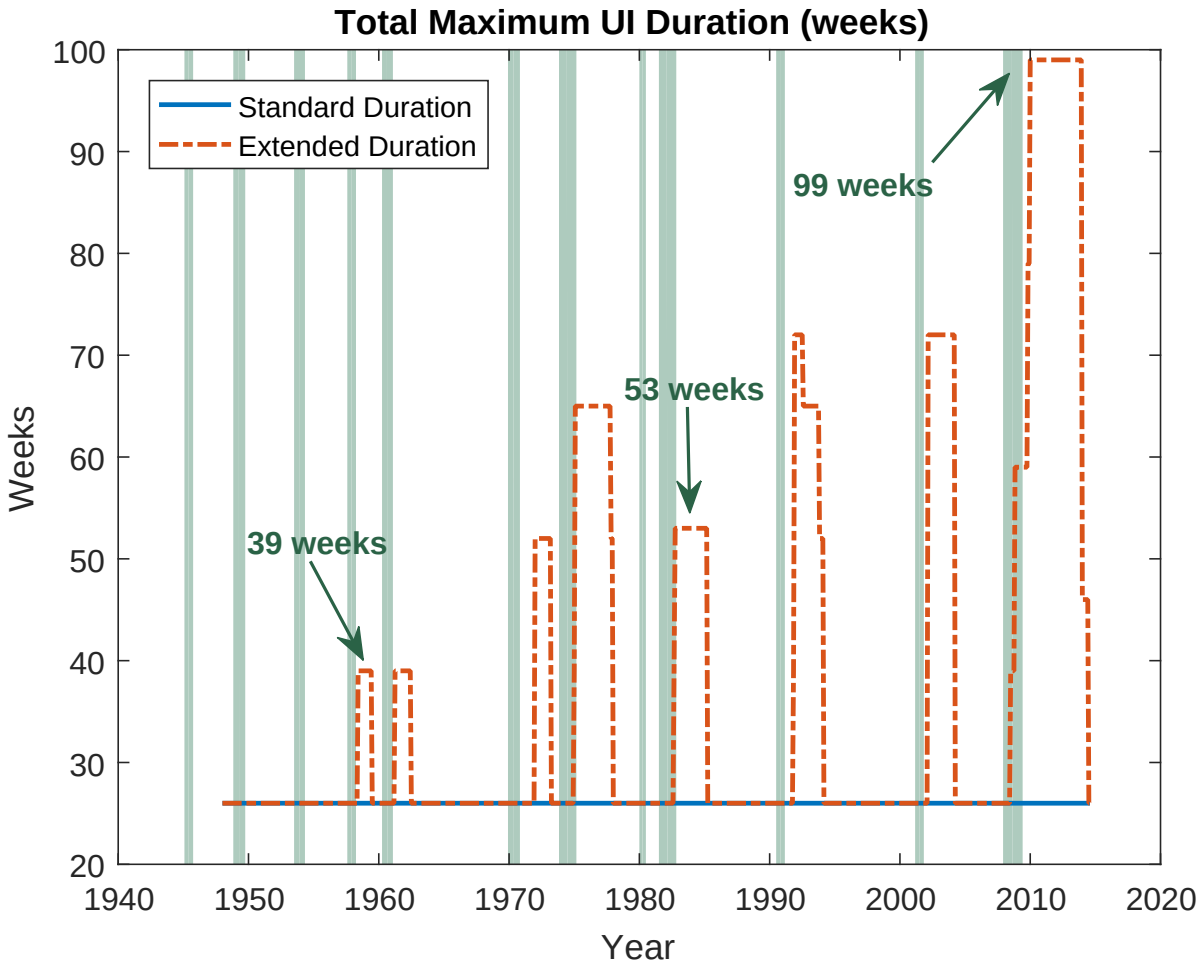
- At maximum, UI recipients can collect the benefits for 26 weeks



Source: ETA

UI Extensions in the US

- In every recession since late 1950s
- More generous over time
- High unemployment triggers extensions



Source: ETA / Shaded areas denote recessions

▸ Automatic ext. ▸ Endogeneity ▸ Micro evidence

Literature

On the rise of long-term unemployment/unemployment duration

- Ahn & Hamilton (2016), Ahn (2016), Hornstein (2012), Kroft, Lange, Notowidigdo & Katz (2014), Elsby, Hobijn, Şahin & Valetta (2011), Ravn & Sterk (2013), Carrillo-Tudela & Visschers (2014), Wiczer (2015), Krueger, Cramer & Cho (2014), Abraham, Haltiwanger, Sandusky & Spletzer (2016)

On the effects of UI extensions

- Farber & Valletta (2011), Fujita (2011), Valletta & Kuang (2010), Rothstein (2011), Hagedorn, Karahan, Manovskii, and Mitman (2015), Barnichon & Figura (2014), Hagedorn, Manovskii, and Mitman (2015), Aaronson, Mazumder, and Schechter (2010), Mitman & Rabinovich (2014), Krusell, Mukoyama and Sahin (2010), Faig, Zhang and Zhang (2012), Nakajima (2012)

Contributions / Preview of Results

Main contributions:

1. UI extensions as a source of long-term unemployment
   - Emphasis on **heterogeneous job finding rates**
2. Construct a general equilibrium search and matching model with
   - Both **macroeconomic + microeconomic effects** of UI extensions
   - **Rational expectations** about the timing of UI extensions

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2. Construct a general equilibrium search and matching model with
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Main results:

1. UI extensions can account for up to 90% of LTU during the Great Recession (non-linear effect)
   - (Limited role from unobserved heterogeneity of workers & true duration dependence)
2. Main UI channel: for LTU $\rightarrow$ job search behaviour (micro)
   For total unemployment $\rightarrow$ job separations (macro)
3. Large overestimation of the UI effect without rational expectations

Model Setup

- Time is discrete, runs forever
- The economy is populated by a continuum of workers of mass one and a large continuum of firms
  - Infinitely-lived, risk neutral, no savings, discount factor $\beta < 1$

Workers:

- Have high or low productivity (type-H or -L)
- 3 employment statuses:
  1. Insured unemployed: u^{UI}
  2. Uninsured unemployed: u^{UU}
  3. Employed: e
- Choosing job search effort: s costing disutility $\nu_j(s)$
  - $\nu_j(\cdot)' > 0, \nu_j(\cdot)'' > 0 \quad$ for $j \in \{e, u\}$

Production and Matches

- A match between a firm and a type-i worker produces:

$$y_{m,i,t} \quad = \quad z_t \times m_t \times \eta_i \quad ; i \in \{H, L\}$$

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  - Redrawn at rate λ for next period ($\rightarrow$ endogenous separation)

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  - Types are permanent

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- **Exogenous match separations** at rate δ
- **Endogenous separations**: when surplus of worker/firm < 0
  - Bad aggregate and/or match-specific productivity shocks
  - UI extensions
  - Job-to-job transitions

Workers and UI Policy

- $b_i(\tilde{m})$ is the UI payment to the insured unemployed, which depends on, as a wage proxy,
  1. $i \in \{H, L\}$ - worker's type
  2. $\tilde{m}$ - the match quality in most recent employment

- Period consumption for a type-i worker

$$c_{it} = \begin{cases} b_i(\tilde{m}) + h & \text{if insured unemployed} \\ h & \text{if uninsured unemployed} \\ w_{it}^{j}(m) & \text{if employed with match quality } m + \text{previous} \\ & \quad \text{employment status } j \in \{e(\tilde{m}), UI(\tilde{m}), UU\} \end{cases}$$

- **UI Eligibility:** Upon job separation, a fraction ψ of employed workers become uninsured, i.e.
  - $Pr(e \to u^{UI} \,|\text{separation}) = 1 - \psi$
  - $Pr(e \to u^{UU} |\text{separation}) = \psi$

UI Duration Policy

- **UI Exhaustion:** The rate u^{UI} (insured) becomes u^{UU} (uninsured) is

$$\phi(u_t) = \begin{cases} \phi_H & \text{when } u_t < \bar{u} \\ \phi_L & \text{when } u_t \geq \bar{u} \end{cases}$$

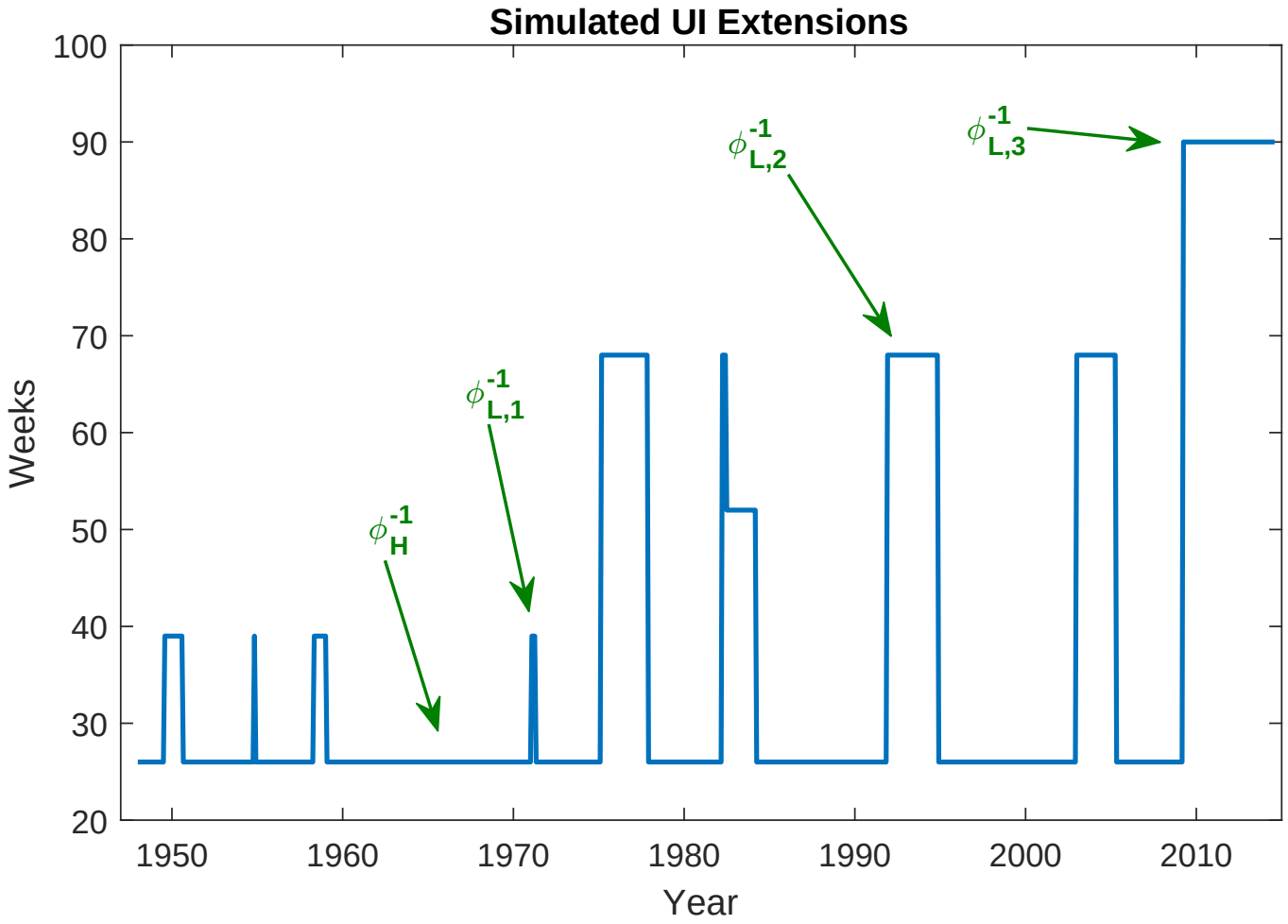
  where $\phi_L < \phi_H$

  - Maximum UI duration is implied by $\phi(\cdot)^{-1}$
  - Extensions occur when $\phi(u_t) = \phi_L$
    - u_t is a state variable: rational expectations

- **UI Finance:** UI budget is balanced each period by taxing producing firms

UI Duration Policy: Example

- Standard maximum UI duration (ϕ_H^{-1}) is constant
- Decreasing $\phi_L \rightarrow$ Increasing UI generosity



Firms / Search Structure

Firms are either

- Matched with one worker: sell output, pay (Nash) wages and lump-sum tax (τ) to finance UI
- Unmatched: post a vacancy at cost κ + Free entry condition

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Random Search

- Meeting function $M(s, v)$: CRS, strictly increasing and concave
  - s = aggregate search intensity
  - v = number of job vacancies
  - Market tightness is $\theta \equiv v/s$
- A worker of (i, j) meets a firm at rate: $p_i^j(\theta) \equiv s_i^j \frac{M}{s} = s_i^j M(1, \theta)$ where $i \in \{H, L\}$ and $j \in \{e(m), UI(\tilde{m}), UU\}$
- An unmatched firm meets a worker at rate: $q(\theta) \equiv \frac{M}{v}$ Timing

Value: Insured Unemployed

- Aggregate state is $\omega \equiv \{z, e_i(m), u_i^{UI}(\tilde{m}), u_i^{UU}\}$
- Value of type-i insured unemployed with benefit $b_i(\tilde{m})$ is

$$
\begin{aligned}
U_i^{UI}(\tilde{m};\omega) = \max_{s_i^{UI}(\tilde{m},\omega)} \quad & b_i(\tilde{m}) + h - \nu_u(s_i^{UI}(\tilde{m},\omega)) \\
& + \beta E_{m'\omega'|\omega}\Bigg[p_i^{UI}(\tilde{m};\omega) \underbrace{W_i^{UI(\tilde{m})+}(m';\omega')}_{\text{value of meeting a firm}} \\
& + (1 - p_i^{UI}(\tilde{m};\omega))\Big((1-\phi(u)) \underbrace{U_i^{UI}(\tilde{m};\omega')}_{\text{insured unemp}} + \phi(u) \underbrace{U_i^{UU}(\omega')}_{\text{uninsured}} \Big)\Bigg]
\end{aligned}
$$

- $W_i^{UI(\tilde{m})+}(m';\omega')$ = value of $u_i^{UI}(\tilde{m})$ meeting a firm =
$$\max\Big\{ \underbrace{W_i^{UI(\tilde{m})}(m';\omega')}_{\text{take job offer}}, \underbrace{(1-\phi(u))(1-\xi)U_i^{UI}(\tilde{m};\omega') + \big(\phi(u) + (1-\phi(u))\xi\big)U_i^{UU}(\omega')}_{\text{remain unemployed}} \Big\}$$

Equilibrium Definition | BE for employed workers | BE for matched firms | BE for unmatched firms

Equilibrium Definition

A recursive competitive equilibrium is characterised by

- **value functions**, $\{W_i^j(m;\omega), U_i^{UI}(m,\omega), U_i^{UU}(\omega), J_i^j(m;\omega), V(\omega)\}$ for $i \in \{H, L\}$ and $j \in \{e(\tilde{m}), UI(\tilde{m}), UU\}$;
- **market tightness** $\theta(\omega)$;
- **search policy** $\{s_i^e(m;\omega), s_i^{UI}(m,\omega), s_i^{UU}(\omega)\}$ for $i \in \{H, L\}$;
- **wage functions** $w^j(m;\omega)$ for $j \in \{e(\tilde{m}), UI(\tilde{m}), UU\}$;

such that, given initial distribution of workers, government's policy $\{\tau(\omega), \phi(\omega)\}$ and law of motion for z:

1. Value functions and the market tightness satisfy the Bellman equations for workers and firms, and the free entry condition
2. Search decisions satisfy the FOCs for optimal search intensity
   ▸ FOCs for search
3. Wage functions satisfy the FOCs for the Nash bargaining rule
4. UI's budget constraint is satisfied
5. The distribution of workers evolves according to the transition equations, consistent with the maximising behaviour of agents
   ▸ transitions ▸ Back to Bellman equation for u^{UI}

Calibration

- I calibrate a subset of 12 parameters to the US economy during 1948-2007 (SMM)

- Other parameters are from the literature
- From Gruber (1997), set $b_i(\tilde{m})$ and h to imply 10%(24%) consumption drop for $u^{UI}(u^{UU})$ at 50% replacement rate
  - u^{UI} receives $b_i(\tilde{m}) + h$
  - u^{UU} receives h

Other fixed parameters

Targeted Moments

Moment	Data	Model	SE
$\mathrm{E}(u)$	0.0583	0.0577	(0.0095)
$\mathrm{E}(\rho_{UE})$	0.4194	0.4286	(0.0198)
$\mathrm{E}(\rho_{EU})$	0.0248	0.0251	(0.0002)
$\mathrm{E}(\rho_{EE})$	0.0320	0.0320	(0.0096)
$\mathrm{E}(u_{dur})$	14.116	13.063	(1.2420)
$\mathrm{E}(u^{UI})$	0.0290	0.0347	(0.0079)
$\mathrm{std}(u)$	0.1454	0.1453	(0.0194)
$\mathrm{std}(\rho_{UE})$	0.0999	0.1402	(0.0138)
$\mathrm{std}(\rho_{EU})$	0.0890	0.0641	(0.0102)
$\mathrm{std}(u_{dur})$	6.9327	6.1954	(0.5806)
$\mathrm{std}(LP)$	0.0131	0.0104	(0.0003)
$\mathrm{corr}(LP, LP_{-1})$	0.7612	0.7593	(0.0181)

- ρ_{UE}: job finding rate // ρ_{EU}: job separation rate // ρ_{EE}: job-to-job transition rate
- u_{dur}: average unemployment duration (weeks)
- $LP = y/(1-u)$: labour productivity (quarterly)

▸ Non targeted moments ▸ Fitted parameters

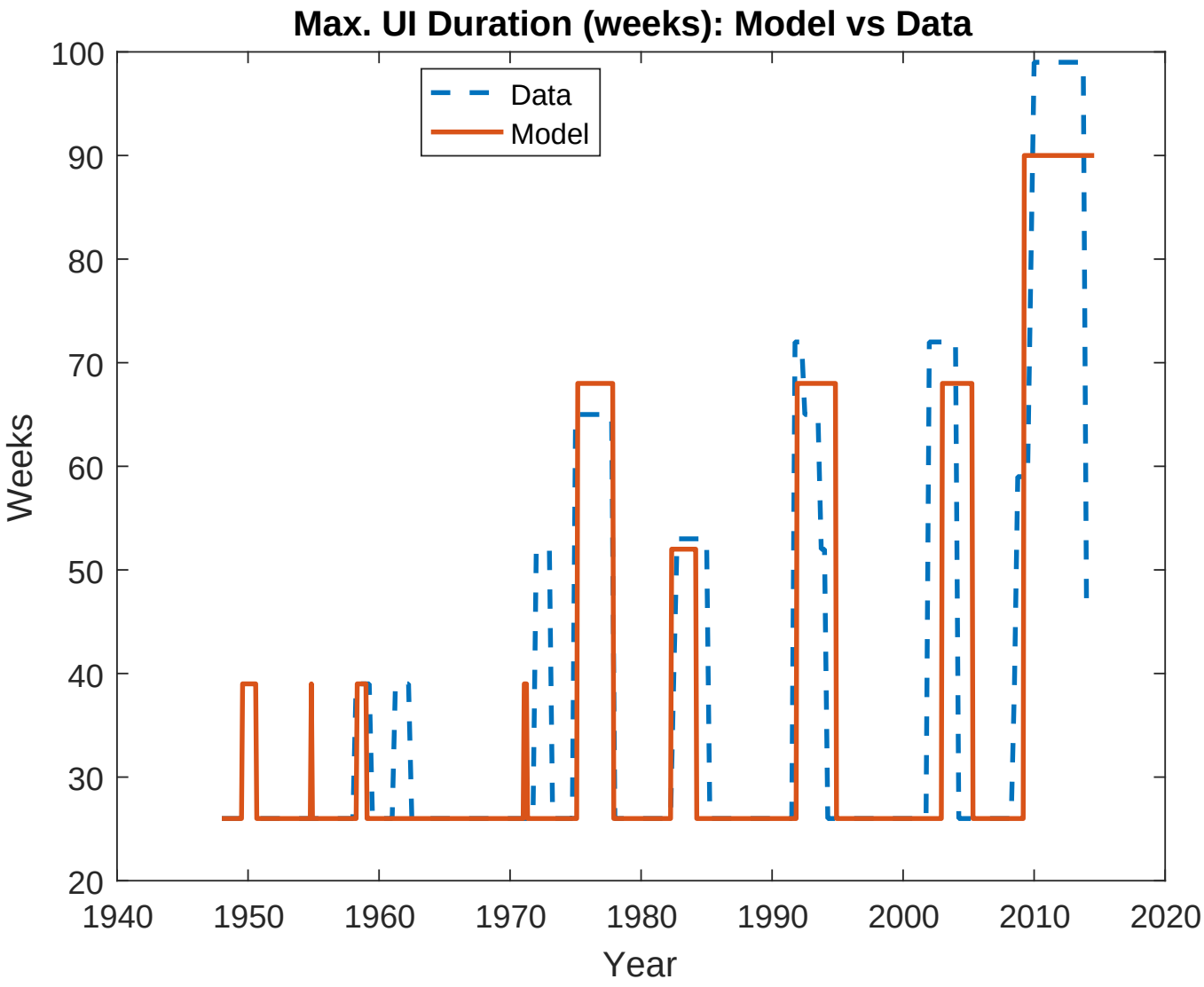
Application to the US

- I apply the model to the US economy by feeding in:

  1. Observed maximum UI durations: Exogenous change in ϕ_L to match data (but timing of extension still endogenous)

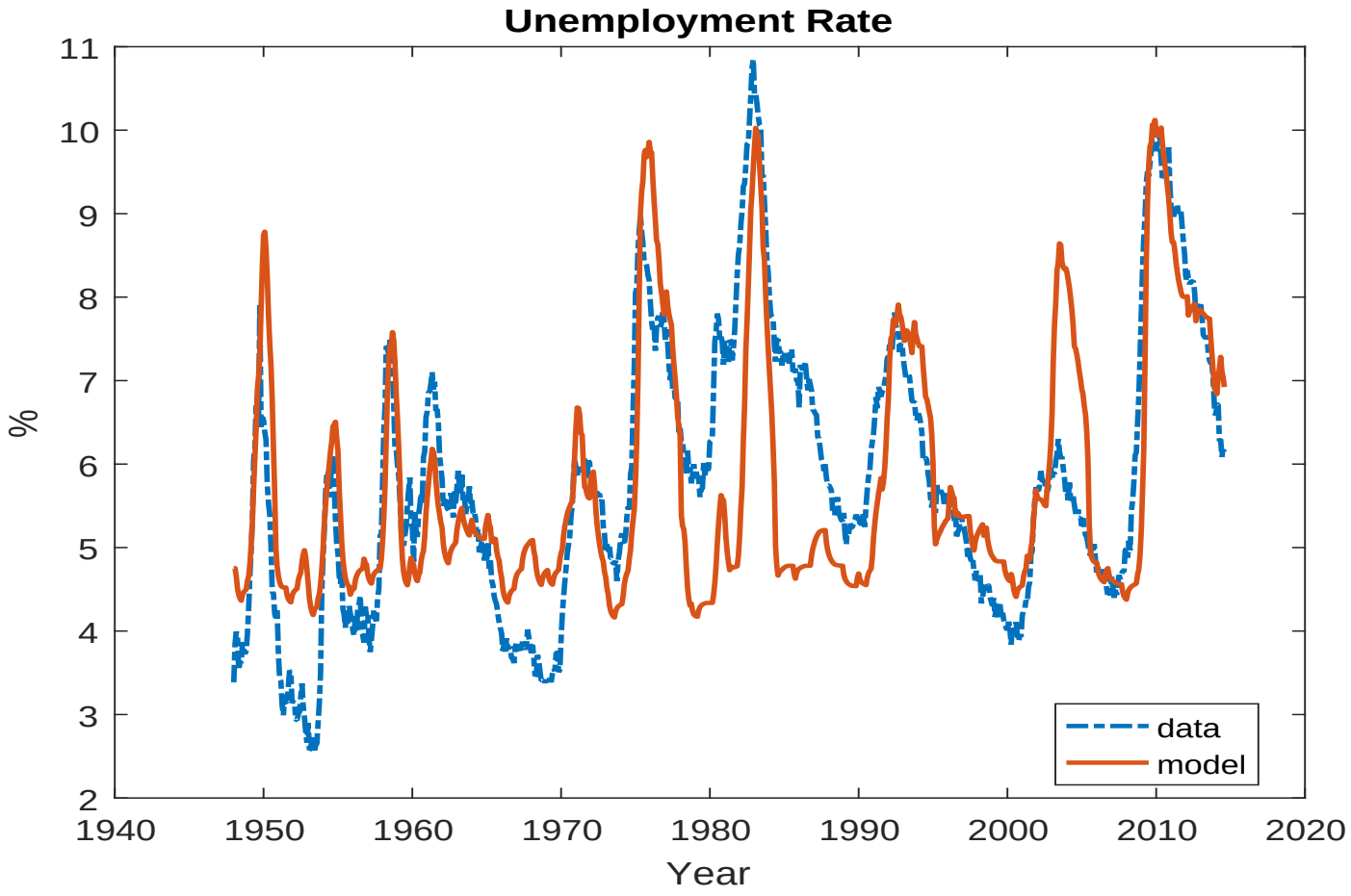
  2. Productivity shocks (z) that match the deviations of output (GDP per capita) from HP trend ▸ z series

UI Extensions



- The model captures well the timing and duration of UI extensions

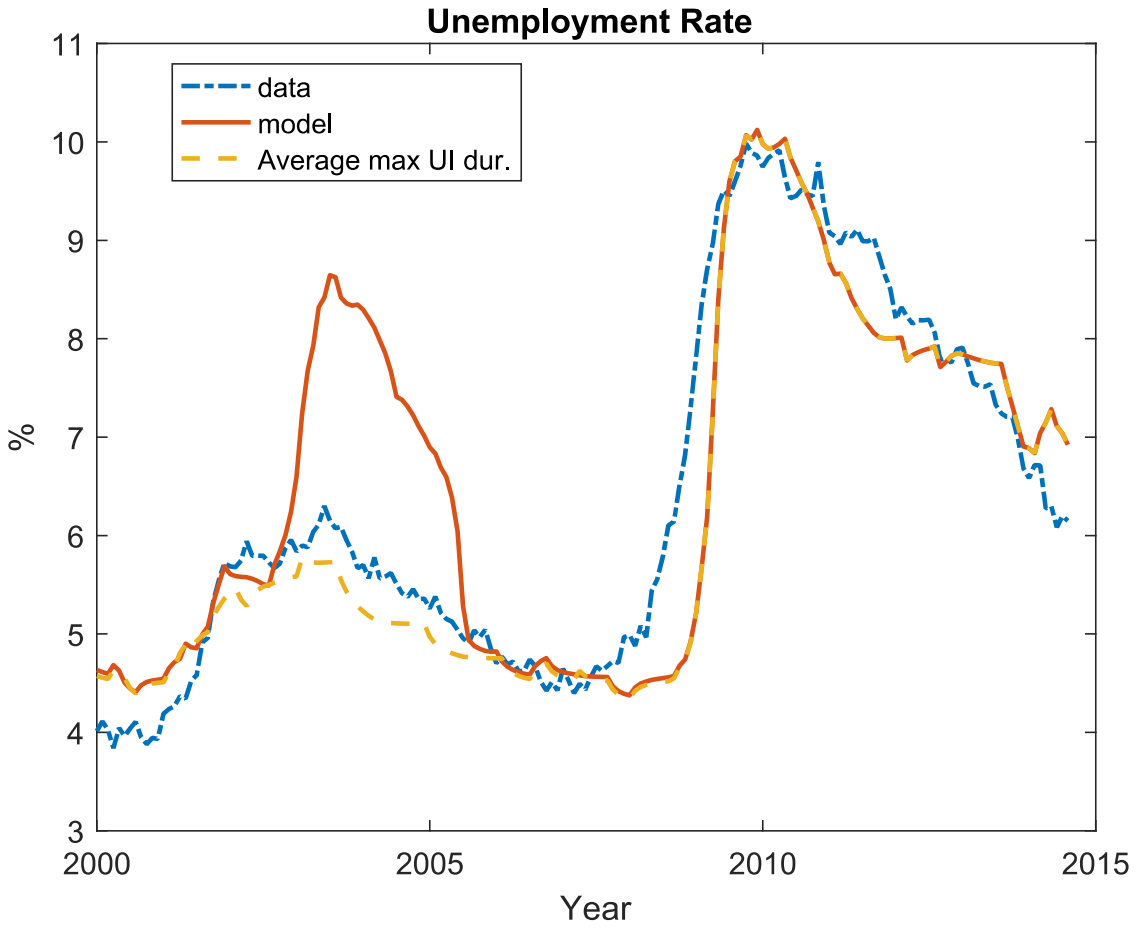
Unemployment Rate



- Model can account for 95% of the observed unemployment during the Great Recession

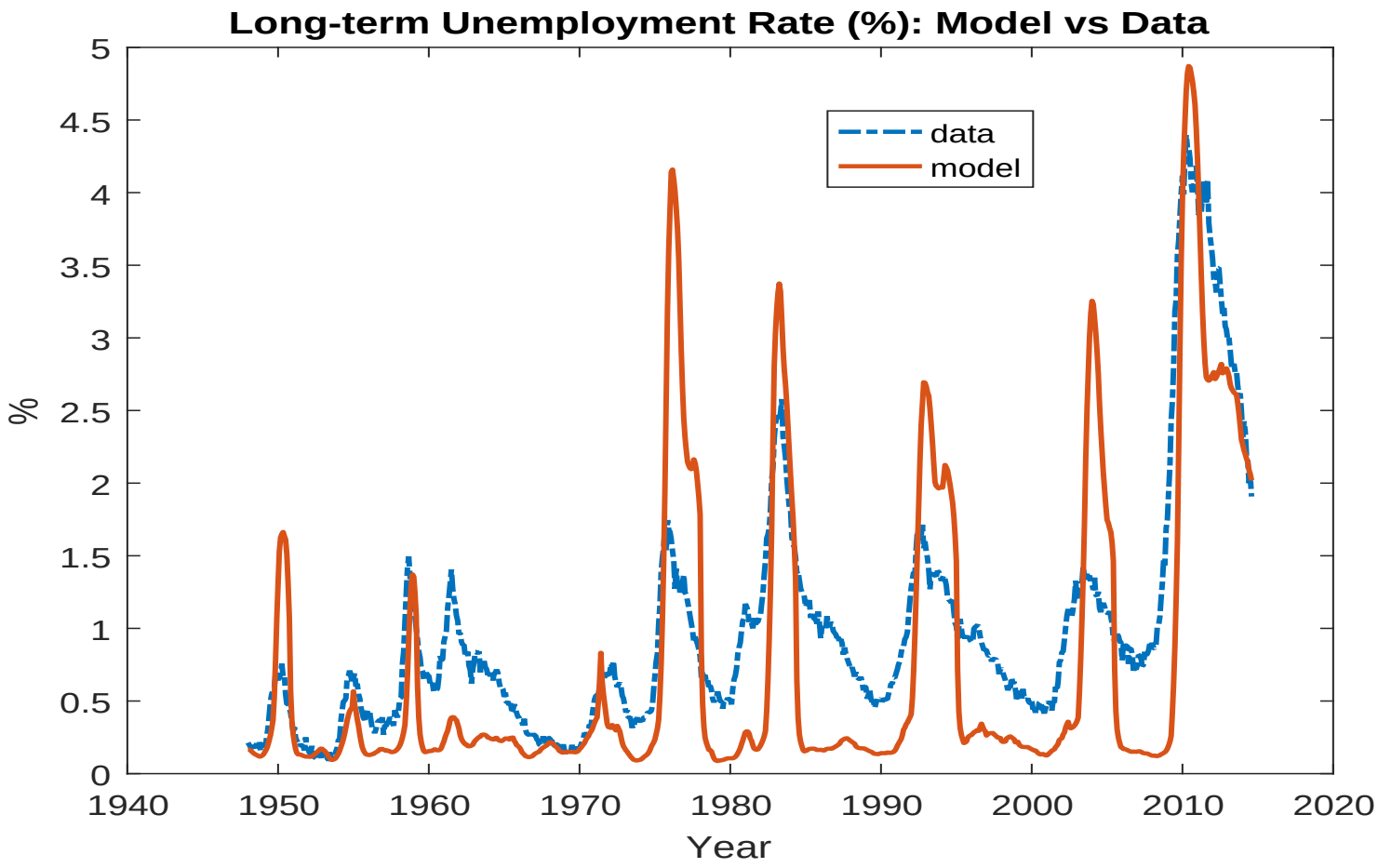
Job findings/job separations | Average unemp. duration | z series | u^{UI} series

Unemployment Rate



- Using max(max UI dur) vs. mean(max UI dur) over US states
- 2000-recession: 5 states implemented all extensions (30-week difference)
- The Great Recession: 49 states implemented (6-week difference)

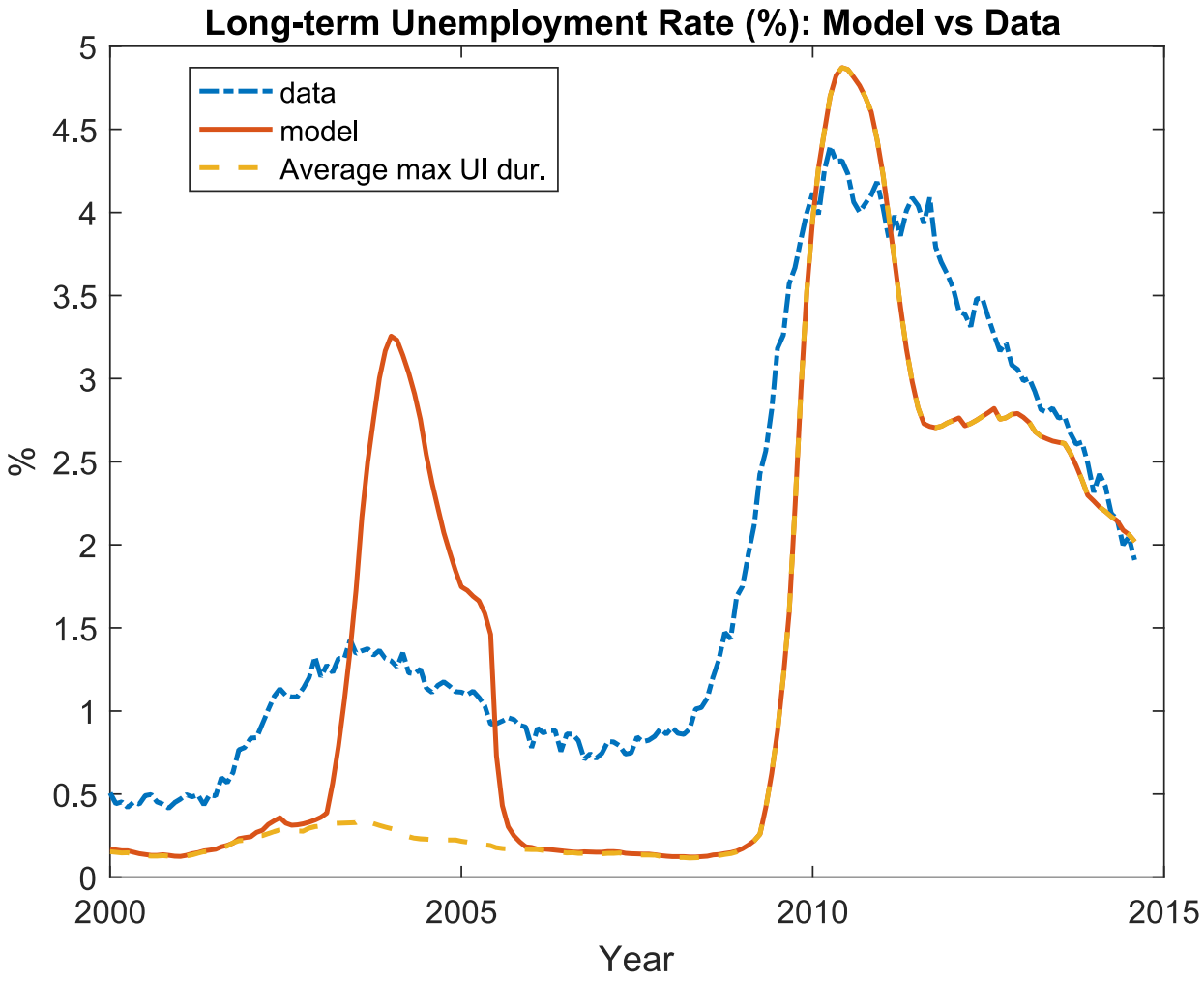
Long-term Unemployment



- Model can account for 87% of the observed LTU during TGR
- But overshooting in some recessions + lacks persistence
  - State implementation of UI extensions
  - Sudden change in job search behaviour

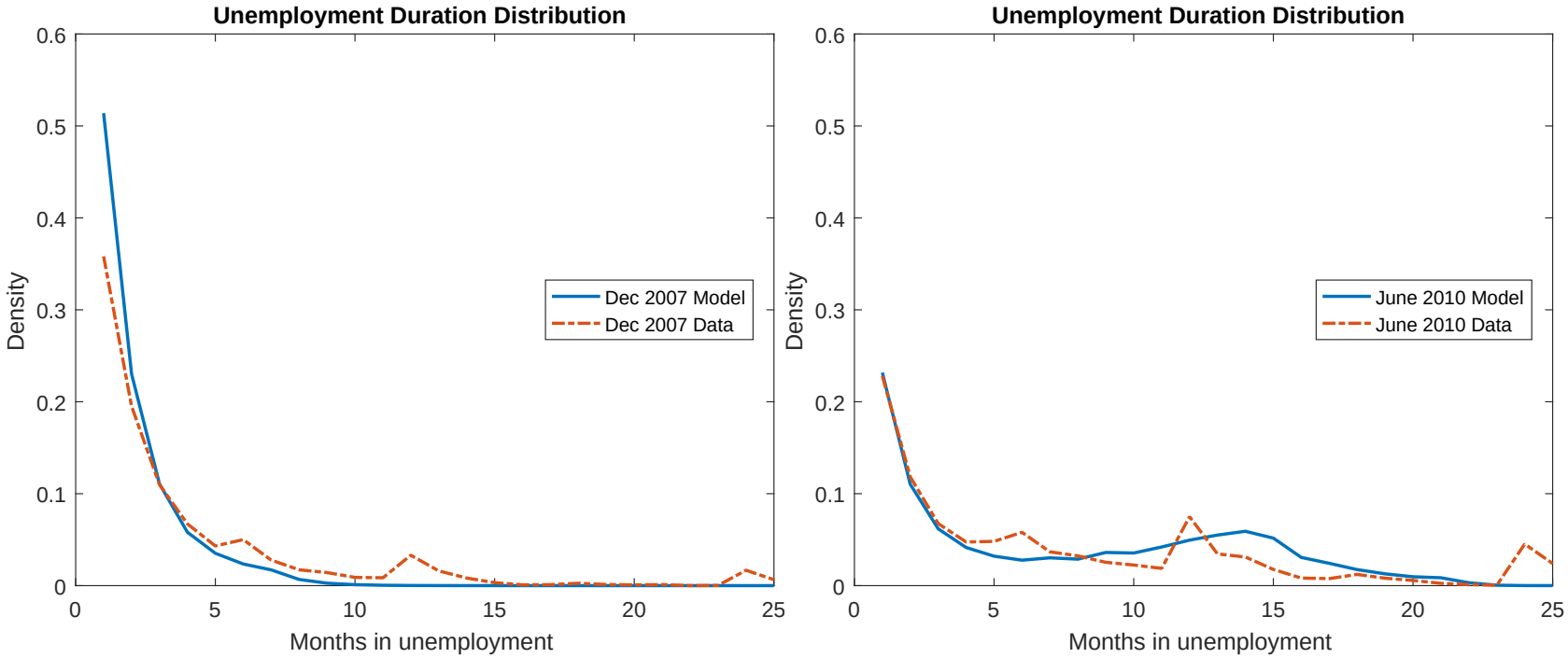
▸ LTU share in total U ▸ Unemploymet shares by 4 duration bins ▸ z series

Long-term Unemployment



- Using max(max UI dur) vs. mean(max UI dur) over US states
- No simulated UI extensions in the early 2000s

Unemployment Duration Distribution: The Great Recession



- Distributions of unemployment durations: Before (left) and after the Great Recession (right)
- Shifting to longer duration bins during the recession

▸ Average unemp. duration

Mechanism: How UI Extensions drive LTU?

Via 3 channels

1. **Job search behaviour**: Workers search less hard
2. **Match formation (job creation)**: Workers become more selective w.r.t. job offers $\rightarrow$ fewer jobs created
3. **Job separation**: Induce employed workers on the margin to return to unemployment

For LTU: most responsive to (1)
For total U: most responsive to (3)

Decomposing LTU - Figure Decomposing U - Figure

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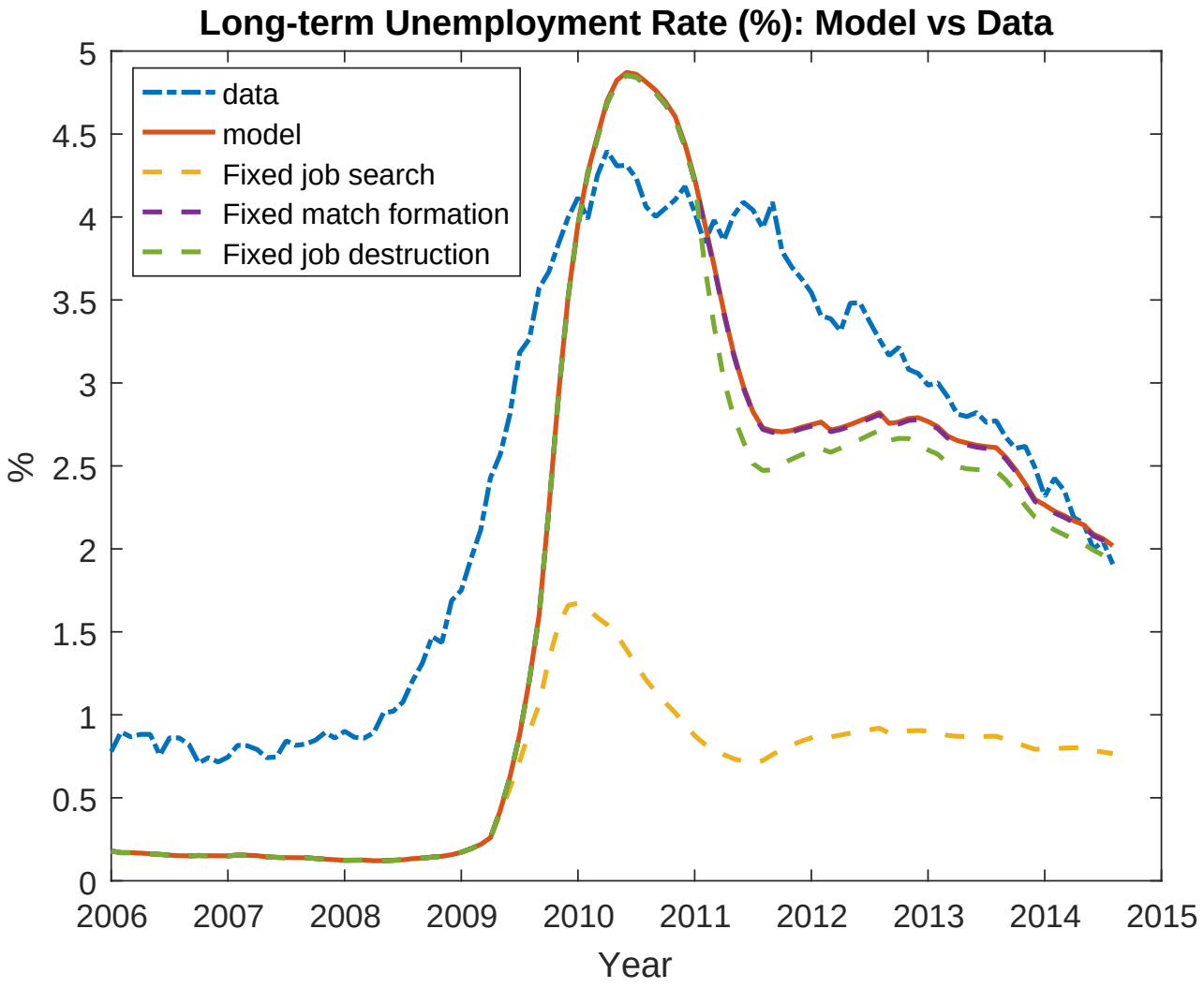
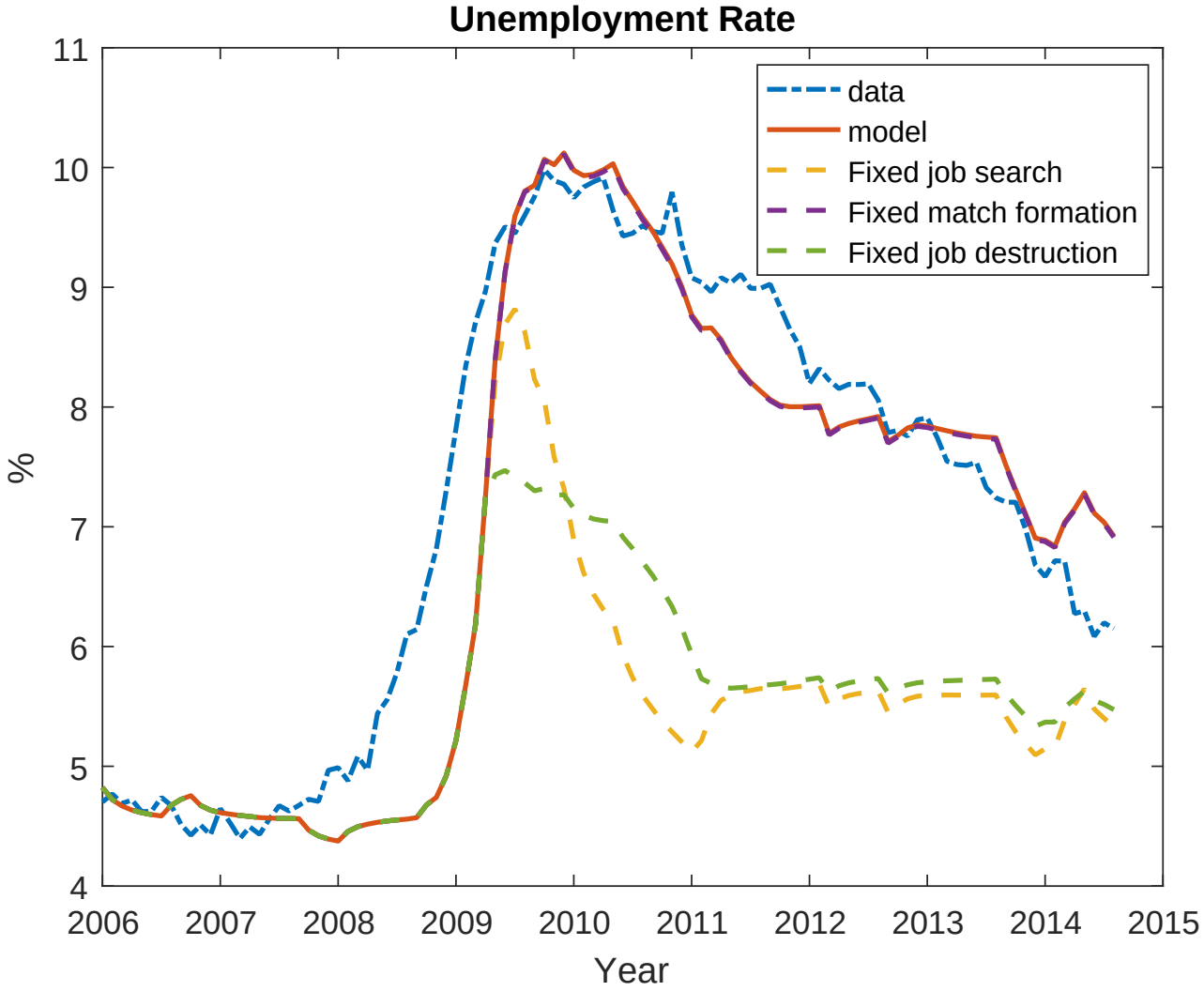


- Fixed x: x doesn't react to UI extensions
- Job search intensity (yellow) most responsive

Decomposing unemployment | Decomposing average unemp. duration | Search policy function

Mechanism: How UI Extensions drive Total U?



- Fixed x: x doesn't react to UI extensions
- Job destruction (green) most responsive

Decomposing average unemp. duration
Job separations: data

Experiment: Eliminating UI Extensions

- **Micro Effects:**

  (1) Direct impact from changing UI exhaustion rate ϕ
  - Now maximum UI duration is only 26 weeks during the Great Recession

  (2) Fixed response of **job search intensity** to UI durations
  - Workers exert job search effort as if UI extensions did not exist

- **Macro Effects:**

  (3) Fixed response of **job separations** to UI durations:
  - Decision to stay in a match not affected by UI extensions

Eliminating UI Extensions in the Great Recession

	Recession Peak	Micro Effect		Micro+Macro Effect
LTU	4.9%	−3.7pp	−4.3pp	−4.4pp
Total U	10.1%	−0.9pp	−1.7pp	−3.4pp
U Duration	38.8	−24.0	−25.9	−26.6

Micro: exhaustion rate and job search intensity // Macro: job separations ▸ table w/data

1. Large UI effect on LTU: account for 90% of the rise
2. Small effect on total unemp.: consistent with existing (micro) estimates: [0.1,1.8] pp. drop
3. Micro and macro effects equally important for total unemp.
4. Non-linearity: smaller effect in previous recession
   - Total U in early 1990s would have been only be 0.7pp lower

Rational Expectations about UI Extensions

What if ϕ is independent of u, i.e. agents perceived UI extensions to last forever (myopic)

- More drastic responses to UI extensions
  - Lower job search effort from the insured unemployed
  - Less likely to form a match
  - More likely to separate from a match

	Data	Baseline Peak	No Rational Exp. Peak
LTU	4.4%	4.9%	7.0%
Total U	10.0%	10.1%	12.2%
U Duration	40.6	38.8	42.0

Note: Periods between October 2009 (peak of US unemp. rate) and June 2014

▸ Extension: Drop in search efficiency ▸ Extension: Unobserved heterogeneity

OJS and UI Extensions

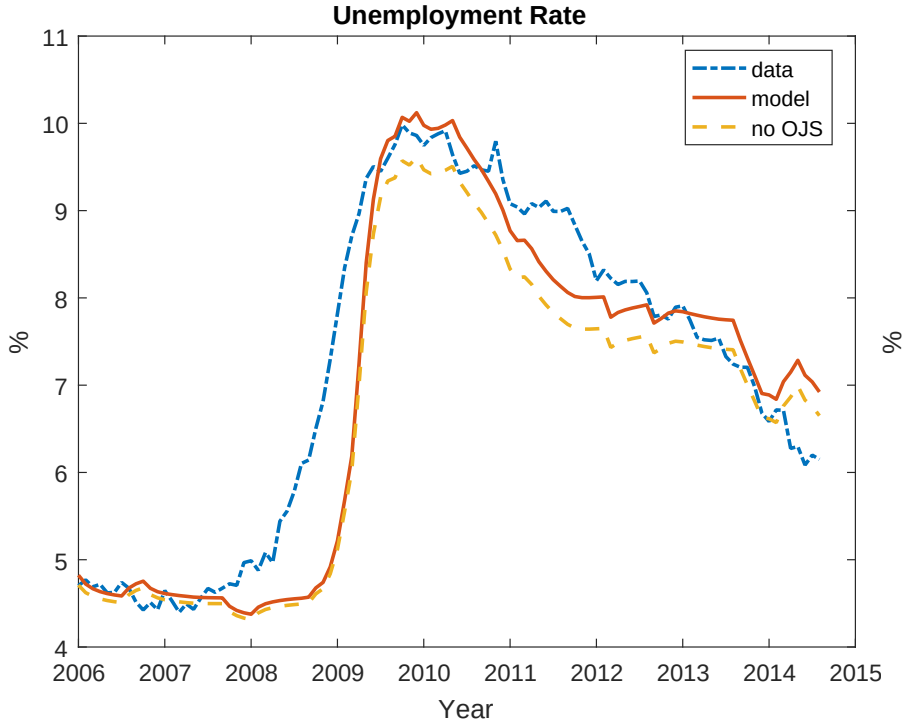
- On-the-job search (OJS) potentially decreases and increases unemployment and its duration

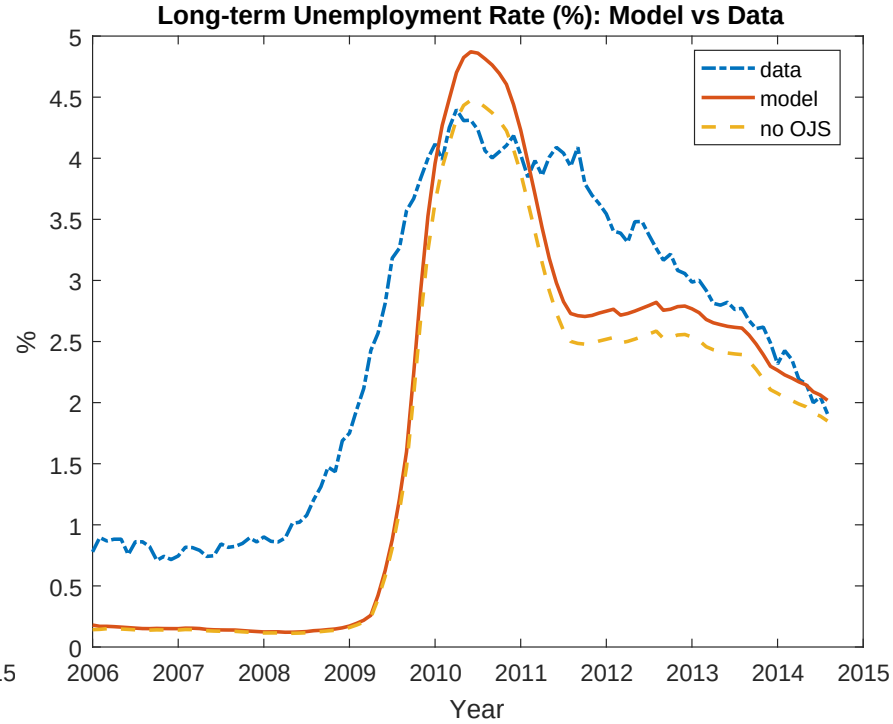
- Decreasing unempl.: OJS increases the value of being employed
  - Induce unemployed workers to take up job offers even when the inital match quality is not great

- Increasing unempl.: OJS allows workers to improve match quality, implying a higher associated UI benefit level
  - Recall that $\uparrow$ benefits $\rightarrow \downarrow$ search efforts, job finding rates

OJS and UI Extensions

- During the Great Recession, OJS contributes to
  - 0.5 pp. increase in total unemployment
  - 0.4 pp. increase in long-term unemployment
  - 1.1 week increase in average unemployment duration
- Almost zero effect outside the recession





Conclusion

- UI extensions account for up to 90% of the rise in long-term unemployment during the Great Recession (non-linear!)
- Mainly from the response of job search intensity of the insured unemployed
- Smaller UI effect on total unemployment: Job separations more important $\rightarrow$ Policy implication/effectiveness

Extensions

- Rational expectations are important for the UI effects
- Limited role for unobserved heterogeneity of workers and true duration dependence in job finding rates
- The role of UI history: UI-exhausted workers and job search behaviour (drop in search efficiency)

Thank you